

HACK WITH INFY SAMPLE QUESTIONS



Sample 1

Today you decided to go to the gym. You currently have E energy. There are N exercises in the gym. Each of these exercises drains Ai amount of energy from your body.

You feel tired if your energy reaches 0 or below. Calculate the minimum number of exercises you have to perform such that you become tired. Every unique exercise can only be performed at most 2 times as others also have to use the machines.

If performing all the exercises does not make you feel tired, return -1.

Parameters:

E :: INTEGER

The first line contains an integer, E, denoting the Energy.

E :: 1 -> 10^5

N :: INTEGER

The next line contains an integer, N, denoting the number of exercises.

N :: 1 -> 10^5

A :: INTEGER ARRAY

Each line i of the N subsequent lines (where 0 ≤ i < N) contains an integer describing the amount of energy drained by ith exercise.

A[i] :: 1 -> 10^5

Case#: 1

Input:

6
2
1
2

Output:

4
E = 6
Do 1st exercise 2 times
Do 2nd exercise 2 times
Hence, total exercise done 4.

Case#: 2

Input:

10
2
1
2

Output:

-1
E = 10
By doing both the exercises 2 times you won't feel tired.
Case#: 3

Input:

2
3
1
5
2

Output:

1
E = 2
Use 3rd exercise 1 time.
Hence, total exercise done 1.



Sample 2

There is a battle between heroes and villains going on. You have M heroes, all of them have the same health H . There are N villains, health of the i th villain is V_i .

When a hero, with health H battles a villain with health V_i , one of the three scenarios can happen:

if $H > V_i$: The villain is defeated and the health of the hero is decreased by V_i

if $H < V_i$: The villain wins, his health is not affected and the hero is no longer able to fight.

if $H = V_i$: Both of them are considered defeated and neither can fight.

The heroes start fighting villains one by one in the same order, first villain 1 then villain 2 and so on. It is might be possible that before defeating all the villains, all the heroes are defeated. Therefore, to ensure the victory of the heroes, you want to remove some villains from the front.

Your task is to find the minimum number of villains you need to remove from the front such that the victory of the heroes is guaranteed.

Note: If in the last battle, both the hero and villain are defeated and no more heroes or villains remain, it would still be considered a victory since all the villains are defeated.

Parameters:

$N :: \text{INTEGER}$

The first line contains an integer, N , denoting the number of villains

$N :: 1 \rightarrow 2 \cdot 10^5$

$M :: \text{INTEGER}$

The next line contains an integer, M , denoting the number of heroes

$M :: 1 \rightarrow 2 \cdot 10^5$

$H :: \text{INTEGER}$

The next line contains an integer, H , denoting the health of each of the heroes

$H :: 1 \rightarrow 10^9$

$\text{array} :: \text{INTEGER ARRAY}$

Each line i of the N subsequent lines (where $0 \leq i < N$) contains an integer describing the health of each of the villains.

$\text{array}[i] :: 1 \rightarrow 10^9$

Case#: 1

Input:

4

4

3

3

1

3

3

Output:

0

[3, 1, 3, 3]. We have 4 heroes will health 3. The heroes 1 will fight villain 1. Both get defeated. The hero 2 fights villain 2. It wins the battle and now his health is 2. He fights the third villain and loses, the villain still has health 3. The hero 3 fights villain 3 and both get defeated. Hero 4 fights villain 4 and both get defeated. So no need to remove any villain.

Case#: 2

Input:

5

3

3

1

2

3

1

1

Output:

0

The fight will take place and hero 1 will defeat villain 1 and 2. Hero 2 will defeat villain 2. Hero 3 will defeat villain 3 and 4

Case#: 3

Input:

5

1

4

1

2

3

1

3

Output:

3

Only 1 hero is present with health 4. Since you can only remove villain from the front, you will have to remove the first 3 villains to ensure victory. The hero can fight the last 2 villain of health 1 and 3 respectively and win the battle.

Sample 3

You need to build a road in a rugged terrain. You know the sea level of each segment of the rugged terrain, i.e. the i -th segment is L_i meters from sea level.

You need to transform the terrain into a strictly downward sloping terrain for the road, i.e. for each i -th segment where $2 \leq i \leq N$, resultant $L_{i-1} > L_i$. In order to do so, you employ a powerful digging team to help you dig and reduce the sea level of the segments. On day D , the team can reduce the sea level for each segment that you scheduled that day by $2D-1$ meters each.

You are allowed to assign the team to dig on multiple segments and/or dig on the same segments for multiple days.

Your task is to find the minimum number of days needed to transform the terrain as per your requirements.

Parameters:

$N :: \text{INTEGER}$

The first line contains an integer, N , denoting the number of elements in L .

$N :: 1 \rightarrow 10^5$

$L :: \text{INTEGER ARRAY}$

Each line i of the N subsequent lines (where $0 < i \leq N$) contains an integer describing L_i , the sea level of the i -th segment.

$L[i] :: -10^9 \rightarrow 10^9$

Case#: 1

Input:

2

3

3

Output:

1

We can dig on the 2nd segment, reducing it from 3-meter sea level to 2. Resulting in $\{3, 2\}$ which is strictly decreasing.

Case#: 2

Input:

2

5

-3

Output:

0

It is already strictly decreasing before start.

Case#: 3

Input:

4

-1

1

1

1

Output:

3

One of the possible way:

On day 1, we can dig on 1st and 4th segment, resulting in $\{-2, 1, 1, 0\}$

On day 2, we can dig on 3rd and 4th segments, resulting in $\{-2, 1, -1, -2\}$

On day 3, we can dig on 2nd, 3rd and 4th segments, resulting in $\{-2, -3, -5, -6\}$



Sample 4

You are given an array of size N. You need to change this array into a mountain. By mountain we mean, the either ends of the array should have equal elements. Then as we move towards the middle from both ends, the next element is just one more than the previous one. So it would have a peak in the middle and decreases if you go towards either end, just like a mountain.

Examples of mountains are [1, 2, 3, 2, 1] or [6, 7, 8, 8, 7, 6]. But the array [1, 2, 4, 2, 1] is not a mountain because from 2 to 4 the difference is 2. The array [1, 2, 3, 1] is also not a mountain because the elements 2 and 3 are not equal from both ends.

You need to find the minimum number of elements that should be changed in order to make the array a mountain. You can make the elements negative or zero as well.

Parameters:

N :: INTEGER

The first line contains an integer, N, denoting the number of elements in array.

N :: 1 -> 10⁵

array :: INTEGER ARRAY

Each line i of the N subsequent lines (where $0 \leq i < N$) contains an integer describing i-th element of array.

array[i] :: 1 -> 10⁶

Case#: 1

Input:

5

1

2

3

4

5

Output:

2

array = [1, 2, 3, 4, 5] . We can change 4 and 5 to make it [1, 2, 3, 2, 1]

Case#: 2

Input:

9

1

1

1

2

3

2

1

1

1

Output:

4

array = [1, 1, 1, 2, 3, 2, 1, 1, 1]. We can change the array to [-1, 0, 1, 2, 3, 2, 1, 0, -1]

Case#: 3

Input:

6

3

3

4

4

5

5

Output:

3

array = [3, 3, 4, 4, 5, 5]. We can chage the array to [2, 3, 4, 4, 3, 2]



Sample 5

You have an interesting string S of length N . It is interesting because you can rearrange the characters of this string in any order. You want to cut this string into some contiguous pieces such that after cutting, all the pieces are equal to one another.

You can't rearrange the characters in the cut pieces or join the pieces together. You want to make the number of pieces as large as possible. What is the maximum number of pieces you can get?

Note: You can observe that you may not want to cut the string at all, therefore the number of pieces is 1. Hence, the answer always exists.

Parameters:

$S :: \text{STRING}$

The first line contains a string, S , denoting the string.

$\text{len}(S) :: 1 \rightarrow 2 * 10^5$

Case#: 1

Input:

zzzzz

Output:

5

You can cut it into 5 pieces "z" + "z" + "z" + "z" + "z"

Case#: 2

Input:

ababcc

Output:

2

Rearrange the string as abcabc. you can cut it into "abc" + "abc", hence the answer is 2

Case#: 3

Input:

abccdcabacda

Output:

2

Rearrange the string as "dcbaca" + "dcbaca", the answer is 2.

5



Sample 6

You are given an array A of size N.
You are allowed to choose at most one pair of elements such that distance (defined as the difference of their indices) is at most K and swap them.

Find the smallest lexicographical array possible after swapping.

Notes:
An array x is lexicographically smaller than an array y if there exists an index i such that $x_i < y_i$, and $x_j = y_j$ for all $0 \leq j < i$. Less formally, at the first index i in which they differ, $x_i < y_i$

Parameters:
N :: INTEGER
The first line contains an integer, N, denoting the number of elements in A.
N :: 1 -> 10^5
A :: INTEGER ARRAY
Each line i of the N subsequent lines (where $0 \leq i < N$) contains an integer describing A[i].
A[i] :: 1 -> 10^5
K :: INTEGER
The next line contains an integer, K, denoting the upper bound on distance of index.

K :: 1 -> N

Case#: 1

Input:

3
2
2
2
1

Output:

2
2
2

Here as all the array values are equal swapping will not change the final result.

Case#: 2

Input:

5
5
4
3

2
1
3
2
4
3
5
1

Output:

Here A=[5,4,3,2,1] K=3, we can swap elements at index 0 and index 3 which makes A=[2,4,3,5,1].

Case#: 3

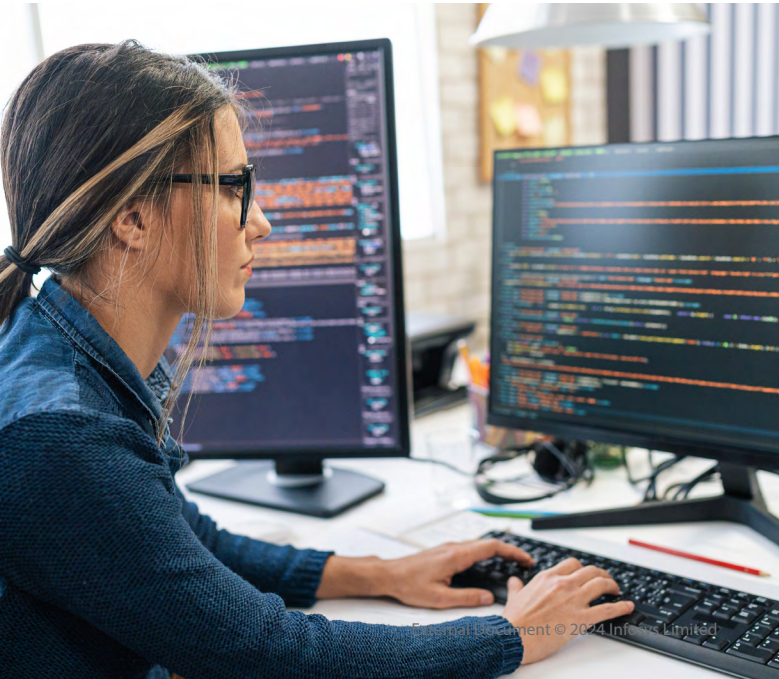
Input:

5
2
1
1
1
1
3

Output:

1
1
1
2
1

Here A=[2,1,1,1,1] K=3, we can swap elements at index 0 and index 3 which makes A=[1,1,1,2,1].



Sample 7

There is a restaurant that offers different types of dishes.

Your friend knows that there are exactly N dishes in the restaurant. The type of the dishes is described by an array Arr . This means that if two elements of Arr have the same value, then they are from the same type.

Your friend wants to eat as many dishes as possible. However, your friend will never order more than one serving of a particular dish.

It is given that the restaurant will not let you order the dishes of the same type more than once. Moreover, if you order A dishes in the restaurant your next order must contain $(2 \cdot A)$ dishes. It is also given that the restaurant does not accept orders containing more than one type of dishes in the same order.

Your friend can start eating with any number of dishes. Find the maximum number of dishes that your friend can eat.

Notes:

Your friend is a foodie and can eat any amount of food that is served to him by the restaurant.

Parameters:

$N :: \text{INTEGER}$

The first line contains an integer, N , denoting the number of elements in Arr .

$N :: 1 \rightarrow 10^5$

$Arr :: \text{INTEGER ARRAY}$

Each line i of the N subsequent lines (where $0 \leq i < N$) contains an integer describing $Arr[i]$.

$Arr[i] :: 1 \rightarrow 10^9$

Case#: 1

Input:

5

1

2

4

2

3

Output:

3

$N=5$

$A=[1,2,4,2,3]$

For example, start with type 1 or 4, then double the amount by eating two of type 2 dishes.

Case#: 2

Input:

7

2

2

1

1

1

1

1

Output:

6

$N=7$

$A=[2,2,1,1,1,1,1]$

Start with eating two dishes of type 2, then eat four dishes of type 1.

Note that you can't start with one dish of type one, then two dishes of type 2, and get back to eat to a dish of size 1 again, your friend cannot eat the same type of dishes multiple times.

Case#: 3

Input:

4

1

1

1

1

Output:

4

$N=4$

$A=[1,1,1,1]$

Your friend can eat all 4 dishes in the first order.

For more information, contact askus@infosys.com

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